OUTBACK ROBOTS

BUILD. LEARN. LEAP AHEAD.

Lesson 3 – Coding

Coding Blossom’s game

HAVIC LAB HUMAN ARTIFICIAL + VIRTUAL INTERACTIVE COGNITION

MONTEBELLO FOUNDATION

Flinders University

Teacher-only slide – Lesson Overview

Lesson Preparation

Before you start:

- Have a go at creating the game with Blossom yourself, if you run into any issues, reach out!
- Familiarise yourself with the Blossom game, it should be easy to understand and encourage the students to play with each other for a few minutes (great metacognitive opportunities, understanding why they give instructions)
- OPTIONAL: run a digital technologies/art class where children build their own Blossom skins in paint, canva or other digital art platform of choice (you are welcome to use the skins we send but children may enjoy creating their own)

Teacher-only slide – Lesson Overview

Lesson 3 - Overview

1. **Lesson 2 Recap** - 5 min
2. **What is coding**– 10 min

   2.1 Introduction

   2.2 Play the guessing game with friends

3. **Code the game in Scratch** – 40 min

   3.1 In-else statements

   3.2 Loops

   3.3 Create your own blocks

   3.4 Code the game

4. **Building Blossom's base** - 25 min
5. **Why coding discussion - optional**
6. **Survey** - 5 min
7. **Summary** - 5 min

Total: 90 min

Teacher-only slide – Lesson Overview

Lesson 3 – Learning objectives

After this lesson, students will be able to:

- explain what coding is and why/where it is important to know,
- design algorithms involving multiple alternatives (branching) and iteration
- implement algorithms using visual programming, involving control structures (such as “if/then”)

Lesson 3 - Student success criteria

- I can explain what coding is and why it is important.
- I can design an algorithm with different outcomes and repeated steps.
- I can use IF statements to control what happens in my program.
- I can use loops to repeat actions in my code.
- I can create and use my own custom blocks with meaningful names.
- I can use variables to store information (like a random number) in Scratch.
- I can explain how robots use sensors, code, and motors to respond to their environment.
- I can describe how computers use binary (0s and 1s) to communicate information.

Recap Lesson 2

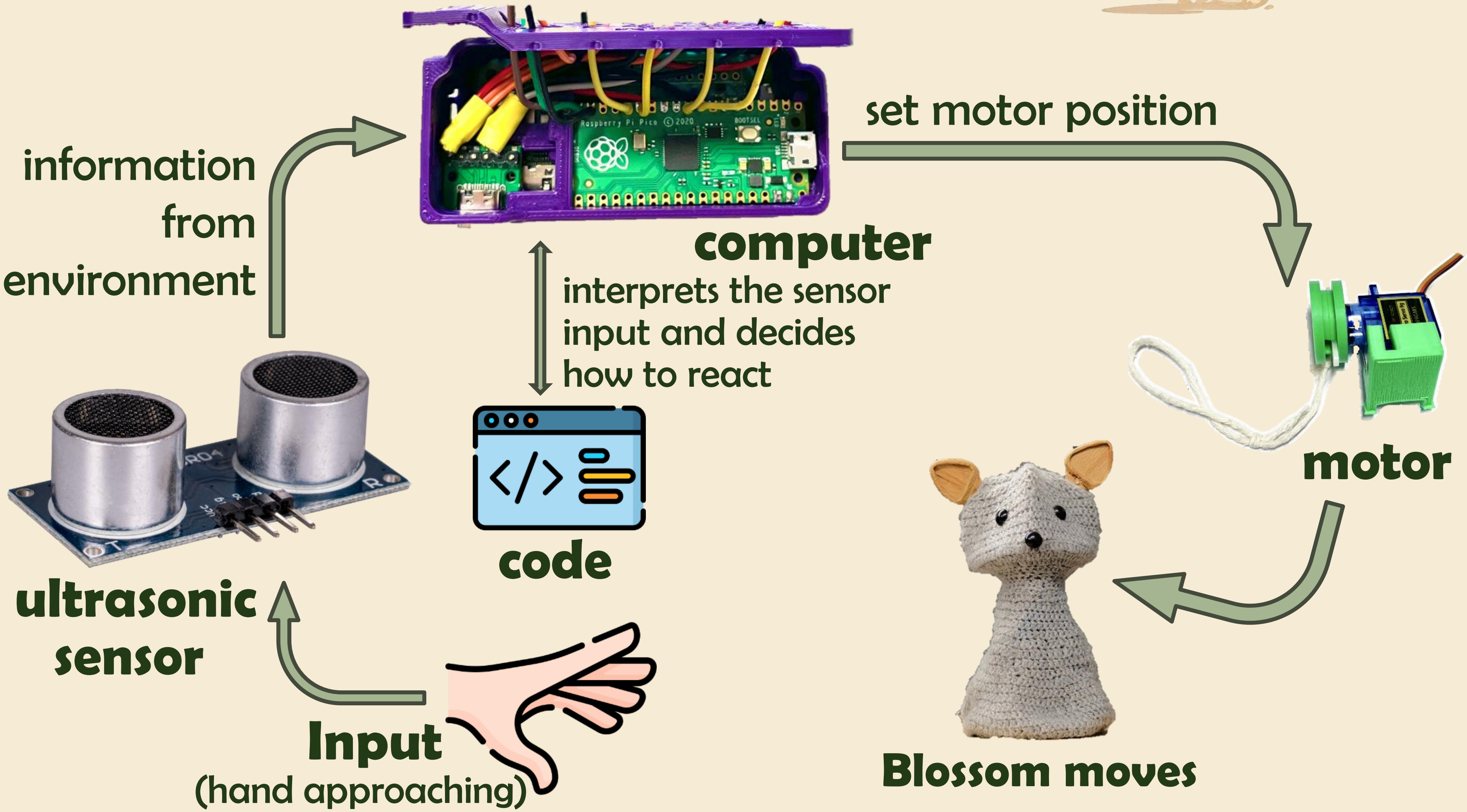
- What is a robot?
- Sensors help robots notice what is happening around them
- Computers use electric signals:
    - ON = 1
    - OFF = 0
- A bit = 1 digit (0 or 1)
- A byte = 8 bits

Teacher-only slide – Instruction Guide

Control processes slide

- The following slide should be familiar
- Explain to the students today we are focusing on the code part of the diagram

Robot Control Process



Teacher-only slide – Instruction Guide

Explaining coding

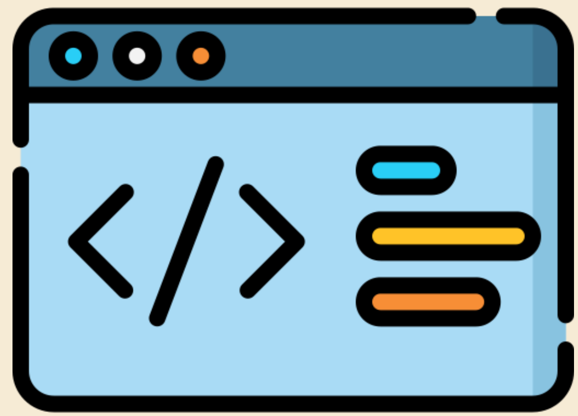
- Coding can seem overwhelming to those that don't have much exposure to it
- Its important to breakdown that barrier and explain coding in a simple way

Explain coding to the students by introducing following concepts:

- *'Code is just a language that robots and computers understand!'*
- *'Like humans, different computers can understand different languages, we will use block coding in these lessons'*
- *'In the code script we build a procedure we want the computer to follow'*
- *'Like us, code has its own dictionary, and if a term we use isn't in their dictionary, we can define it for them'*
- Explain that block-based coding is a good place to start, because we know the computer understands each block, anything inside white bubbles we can change to achieve what we want out of the code

Introduction to coding

- Code = instructions a computer follows exactly
- Like a recipe — miss a step and it goes wrong!
- Block coding = great starting point 🧩



code

Teacher-only slide – Instruction Guide

Blossom's Game

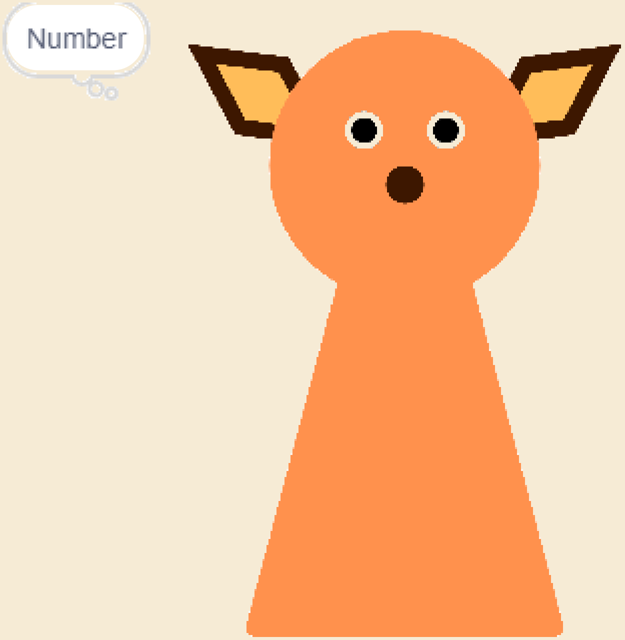
- Blossom's game is quite simple
- It will think of a random number, say between 1 and 20, and we input guesses until we get it right
- We will code Blossom's animation to speak, nod and shake to interact with the students
- We could add competitions for students such as who can get the number in the least guesses etc. to make it fun and competitive

Blossom's Guessing Game

- We are going to code a virtual Blossom (on our screen) to play a game with us

- The goal is to create a game where we guess the number Blossom is thinking of

- Play the game with a friend - "I'm thinking of a number between.."



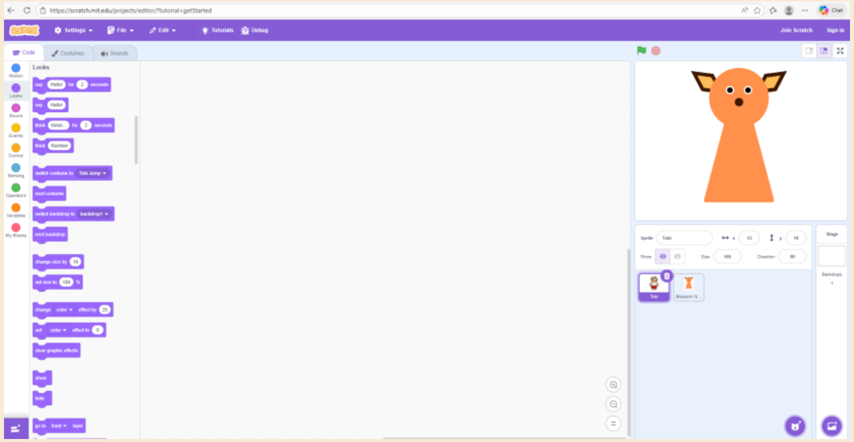
Teacher-only slide – Instruction Guide

Intro to scratch

- Now would be a good time to share your screen
- Open the link on the following slide
- Create a new project
- Walk the children through how to upload their Sprites
- *"Sprites are the little characters that we get to code!*

Intro to Scratch

- Open Scratch and hit create! Scratch - Imagine, Program, Share
- The characters in the top right animation are called our ‘Sprites’
- We can upload our own!



Creating Blossom Guessing Game

- All code needs a place to start!
  - In the EVENTS section we will see many blocks we can choose to begin our code, pick one and drag it to our script

- Underneath, we need to tell the program exactly what to do when your program starts

Creating Blossom Guessing Game

- First, we need Blossom to think of a number, in the operators section you will find this block pick random 1 to 10
    - We can change the numbers to be anything we like

- Once Blossom has thought of a random number, we can then ask the question ask Guess a number between 1 and 20! and wait

Creating Blossom Guessing Game

- At the beginning of the game, we need Blossom to think of a number, in the operators section you will find this block

  - We can change the numbers to be anything we like

- We can make this number a variable, call it something meaningful for when we use it later

Teacher-only slide – Instruction Guide

Introducing IF STATEMENTS

- *"If statements are exactly as they sound, we can code our computers to do something, only IF another event occurs. In block coding, we but that event inside the hexagon next to if. We then place the instructions on what we want it to do between the top and bottom of the block."*
- **Notes: Get students to think of IF STATEMENTS they have in every day life...IF we see the fridge open, we might close it! The students will have 3 if statements inside their code, being their guess is greater than, lower than or equal to Blossoms number. They will provide Blossoms response for each within.**

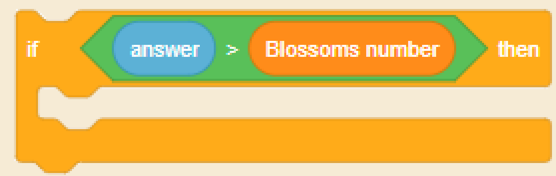
Introducing IF statements

- **IF something happens → THEN do this**
- 🚪 IF the fridge is open → close it
- ☔ IF it's raining → bring an umbrella
- **Find the IF block in the CONTROL section!**
- Inside the hexagon we put our EVENT or condition
- Underneath we put our ACTION (what we want to do)

Our first IF STATEMENT

- Let's start with our first event, if our guess is greater than Blossoms number (you can find the green hexagon inside operators)

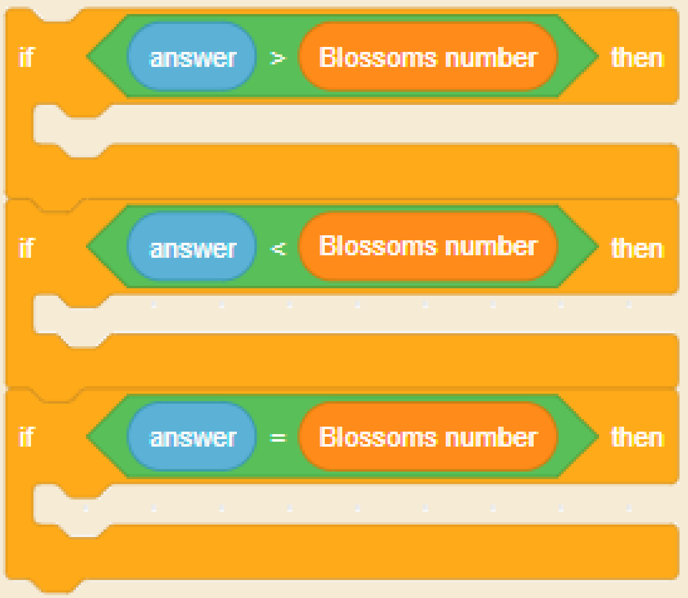
- Clip this inside the hexagon



- Inside the IF block, tell Blossom what you want it to do, this is the ACTION
    - Inside looks we can make Blossom say something (higher or lower)

IF STATEMENTS

- Repeat this for the 3 different events, our number can be less than, greater than or equal to Blossom's



Teacher-only slide – Instruction Guide

Introducing loops

- *"Loops are a stack of code that we want to repeat, this could be forever, until an event happens or a certain number of times. Sometimes it's easier to tell someone to repeat an action 10 times, like throw and catch a ball, rather than telling them to throw the ball, catch the ball, throw the ball, etc."*
- **Note: encourage to think about loops that may be important for the game, repeat the question until students guess correctly, and head nodding and shaking are great examples of loops we will include.**

Introducing loops

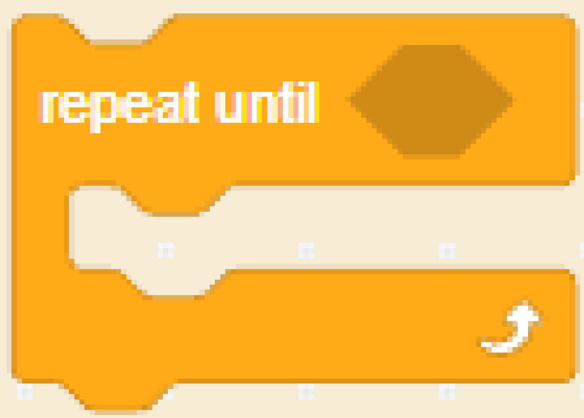
- **A loop repeats code — forever, or until told to stop**

- ⚾ Throw, catch, throw, catch… = loop!
- 🤖 Blossom keeps asking until you guess correctly

- **Use “repeat until” — what event stops it?**

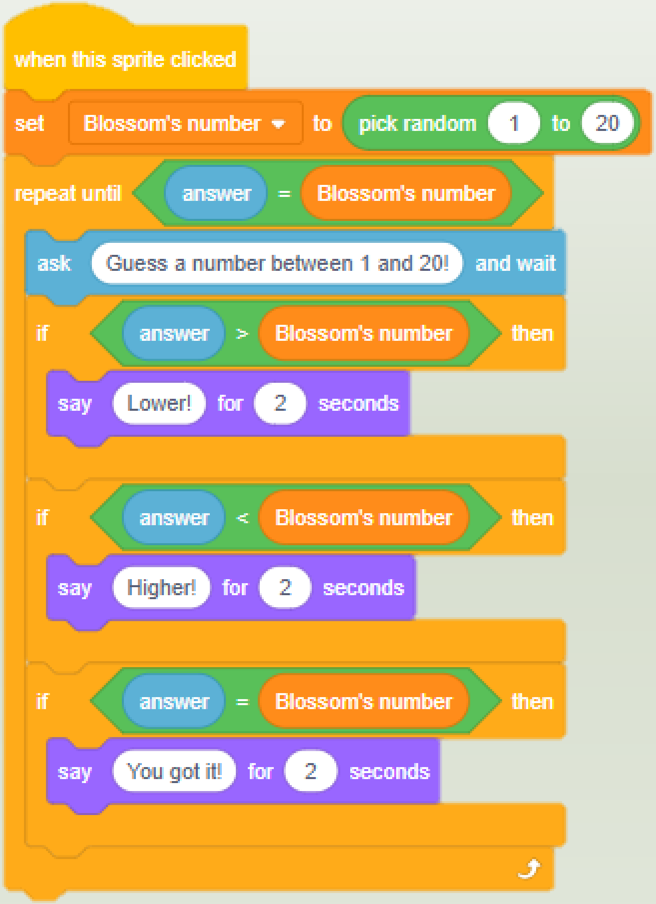
Loops

- When you play the game with your friend, think about when the game stops, and put it into the hexagon!
- We will want all of our code inside the repeat until block!



Creating Blossom Guessing Game- HINT

- **Instruction note: Your code should look something like this. Again, we can play around with all the bubbles in white to better suit our game.**
- **Sample script**: *"Should Blossom speak for longer or would you prefer it to say different things?"*



Building Blossom's Spine

Teacher-only slide – Content Extension

Coding discussion

- Have a discussion with the students about future occupations or daily tasks that may be easier with coding
- Coding is like a problem-solving superpower, what problem would you use coding to help solve
- Even if you're job doesn't need coding skills, it can teach us how to break problems down into smaller steps. When can this be useful?

Survey

Outback Robots Program
DATE: ________________

END-OF-LESSON STUDENT SURVEY

Please rate each statement. There are no right or wrong answers, only your opinion matters.

Section A: Your experience in Today's Lesson

Statement	1	2	3	4	5
1. **Enjoyment.** I enjoyed today's lesson.	○ Strongly Disagree	○ Disagree	○ Not Sure	○ Agree	○ Strongly Agree
2. **Interest.** I found the lesson interesting.	○ Strongly Disagree	○ Disagree	○ Not Sure	○ Agree	○ Strongly Agree
3. **Ability to Follow.** I was able to follow along with what we were learning.	○ Strongly Disagree	○ Disagree	○ Not Sure	○ Agree	○ Strongly Agree
4. **Learning.** I learned something new today.	○ Strongly Disagree	○ Disagree	○ Not Sure	○ Agree	○ Strongly Agree
5. **Difficulty.** The activity was the right level of difficulty for me.	○ Strongly Disagree	○ Disagree	○ Not Sure	○ Agree	○ Strongly Agree
6. **Performance.** I think I was quite good at the activity during the lesson.	○ Strongly Disagree	○ Disagree	○ Not Sure	○ Agree	○ Strongly Agree
7. **Preparation.** I had the information that I needed to complete the activity.	○ Strongly Disagree	○ Disagree	○ Not Sure	○ Agree	○ Strongly Agree
8. **Importance.** I believe that the activity is important.	○ Strongly Disagree	○ Disagree	○ Not Sure	○ Agree	○ Strongly Agree
9. **Confidence.** I felt confident doing the activity.	○ Strongly Disagree	○ Disagree	○ Not Sure	○ Agree	○ Strongly Agree

Section B: Short answers

1. What did you like or dislike during the lesson?

Summary

- Name an example of an IF statement in day-to-day life, what is the EVENT and what is the ACTION
- Why do we use loops?
- What is an advantage of creating our own blocks?